

Package ‘rENM.core’

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Type Package

Title Shared utilities for the rENM framework

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Description The rENM framework is a modular system for reconstructing and analyzing long-term ecological niche dynamics using time-indexed environmental data and species occurrence records. Its packages support climate-based niche modeling, temporal analysis of climatic suitability, and reproducible workflows for studying ecological change.

Depends R (>= 4.1.0)

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URL <https://github.com/rENM-Framework/rENM.core>

BugReports <https://github.com/rENM-Framework/rENM.core/issues>

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get_species_info	<i>Look up species metadata by four-letter alpha code</i>
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Description

Reads the project species table (data/_species.csv) and returns a one-row data frame of standardized metadata for the requested species. Matching is case-insensitive. Column names in the source file are normalized before lookup, so minor variations (e.g., ALPHA.CODE vs SPECIESCODE) are handled automatically.

Usage

```
get_species_info(alpha_code, project_dir = NULL)
```

Arguments

alpha_code	Character scalar. Four-letter species alpha code (e.g., "CASP"). Case-insensitive; leading/trailing whitespace is trimmed.
project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Details

Alpha-code column detection

The function searches for the alpha-code column by normalizing all column names to uppercase with punctuation removed, then matching against these candidates in order: ALPHACODE, ALPHA, SPECIESCODE, SPSC, BANDINGCODE. The first match wins. If none is found, the function errors with the list of available columns.

Output column standardization

Regardless of source column names, the returned data frame always uses these names: COMMON.NAME, SCIENTIFIC.NAME, EBD.RECORDS, EBD.RANGE, GAP.RANGE. The same normalization logic is used to locate these columns in the source file.

Value

A one-row data frame with columns COMMON.NAME, SCIENTIFIC.NAME, EBD.RECORDS, EBD.RANGE, and GAP.RANGE. Row names are reset to NULL.

See Also

[rENM_project_dir](#), [show_species](#)

Examples

```
## Not run:
info <- get_species_info("CASP")
info <- get_species_info("CASP", project_dir = "/projects/rENM")

## End(Not run)
```

rENM_project_dir	<i>Resolve and validate the rENM project root directory</i>
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Description

Returns a normalized absolute path to the rENM project root, resolving it from the first available source in this precedence order: the explicit project_dir argument, the rENM.project_dir R option, or the RENM_PROJECT_DIR environment variable. Errors with an actionable message if no valid directory can be found.

Usage

```
rENM_project_dir(project_dir = NULL)
```

Arguments

project_dir	Character. Path to the rENM project root. Takes precedence over the R option and environment variable. If NULL (default), the option and environment variable are tried in order.
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Details

This function is the single point of filesystem configuration for the rENM framework. All other functions that access the project root call this one rather than resolving paths independently, keeping hard-coded paths out of the codebase and making scripts portable and CRAN-compliant.

A candidate path is accepted only if it is a non-empty scalar character string that expands without error (path.expand) and refers to a directory that already exists (normalizePath(..., mustWork = TRUE)). A candidate that fails any check is silently skipped and the next source is tried.

Configuring the project directory

Explicit argument (recommended for scripts and tests) Pass the path directly: rENM_project_dir("/projects/rENM")

R option (session-level convenience) options(rENM.project_dir = "~/rENM")

Environment variable (persistent across sessions) Add RENM_PROJECT_DIR=~/rENM to ~/.Renviron, or call Sys.setenv(RENM_PROJECT_DIR = "~/rENM") for the current session.

Value

Character scalar. Normalized absolute path to an existing directory.

See Also

[get_species_info](#), [show_species](#), [show_variables](#)

Examples

```
## Not run:
# Explicit path – best for reproducible scripts and tests
rENM_project_dir("/projects/rENM")

# Session option
options(rENM.project_dir = "~/rENM")
```

```

rENM_project_dir()

# Environment variable
Sys.setenv(RENM_PROJECT_DIR = "~/rENM")
rENM_project_dir()

## End(Not run)

```

show_species	<i>Print the project species table to the console</i>
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Description

Reads data/_species.csv from the rENM project directory and prints its contents with a timestamped header. Intended as a lightweight inspection utility for verifying species definitions before running the pipeline.

Usage

```
show_species(project_dir = NULL)
```

Arguments

project_dir Character. Path to the rENM project root. If NULL (default), resolved via [rENM_project_dir](#) (argument, rENM.project_dir option, RENM_PROJECT_DIR environment variable).

Value

Invisibly returns the species data frame. Called primarily for the side effect of printing a formatted table and header to the console.

See Also

[rENM_project_dir](#), [get_species_info](#), [show_variables](#)

Examples

```

## Not run:
show_species()
show_species(project_dir = "/projects/rENM")

## End(Not run)

```

show_variables	<i>Print the project variables table to the console</i>
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Description

Reads `data/_variables.csv` from the rENM project directory and prints its contents with a timestamped header. Intended as a lightweight inspection utility for verifying environmental variable definitions before running the pipeline.

Usage

```
show_variables(project_dir = NULL)
```

Arguments

project_dir	Character. Path to the rENM project root. If NULL (default), resolved via rENM_project_dir (argument, <code>rENM.project_dir</code> option, <code>RENMPROJECTDIR</code> environment variable).
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Value

Invisibly returns the variables data frame. Called primarily for the side effect of printing a formatted table and header to the console.

See Also

[rENM_project_dir](#), [show_species](#)

Examples

```
## Not run:
show_variables()
show_variables(project_dir = "/projects/rENM")

## End(Not run)
```

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